

- μ_i = expected return of i -th asset
- ρ_{ij} = correlation between asset i and j .
($-1 \leq \rho_{ij} \leq 1$)
- σ_i = standard deviation of asset i -th
- R = desired expected return

Decision variables

w_i = Amount of i -asset invested
($w_i \geq 0$) Do not allow shorting

$$\text{minimize: } \sum_i \sum_j s_i \rho_{ij} \sigma_j w_i w_j \quad (1)$$

$$\text{such that } R = \sum_i w_i \mu_i \quad (2)$$

$$\sum_i w_i = 1 \quad (3)$$

$$0 \leq w_i \leq 1 \quad \forall i \in \{1, \dots, N\} \quad (4)$$

minimize : the sum of the risk times the correlation times amount invested

$$\text{risk} = \sigma_i$$

$$\text{correlation} = \rho_{ij}$$

$$\text{amount invested} = w_i$$

such that : the return which is:
the sum of each investment times its expected return
expected return = μ_i

Investing $N=2$
and each asset is

between 0 and 1

Let σ_{ij} (covariance between assets)

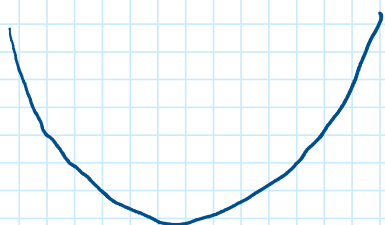
$$\sigma_{ij} = s_i s_j \rho_{ij} \quad \text{then we can}$$

$$\text{minimize } \sum_i \sum_j \sigma_{ij} w_i w_j$$

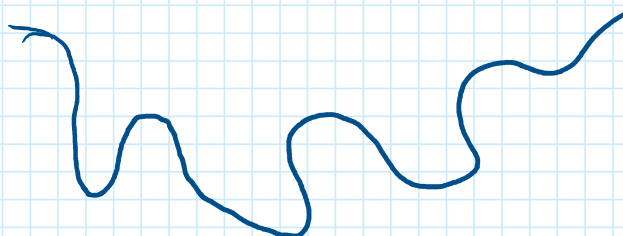
Since all covariance matrices are positive semidefinite the solution is unique.

In addition, even though this is a non linear problem the SDP σ_{ij} allows efficient solver implementation

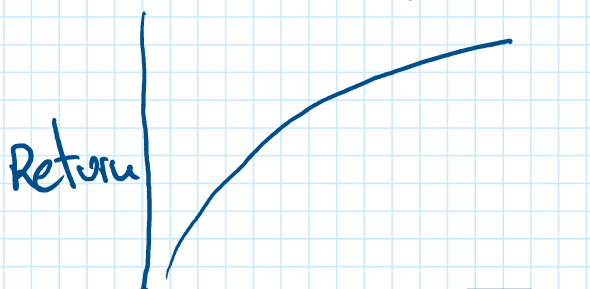
Matrix SDP

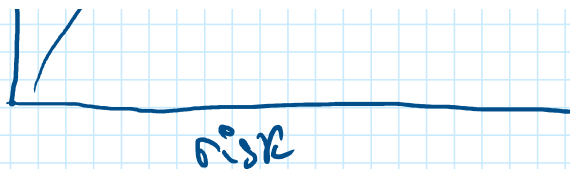


Matrix No-SDP



The formulation above built an efficient frontier (EF), a smooth non decreasing curve which gives the best possible trade-off of risk vs return





the rest of the paper adds complexity because it supposes scenarios where we work when S_t is invested to any asset, the portfolio is rebalancing, the rebalancing takes money...

The solver used OSQP:

OSQP is an open-source wrapper for OSQP written in C.

It always solve the same kind of problem

$$\text{minimize}_x \quad \frac{1}{2} x^T P x + q^T x$$

$$\text{such that} \quad l \leq x \leq u$$

where $x \in \mathbb{R}^n$ is the optimization variable and $P \in \mathbb{S}_+^n$ is a PSD matrix

Therefore, the first step is translating Markowitz's formulation to OSQP

Markowitz	OSQP	Value
$\min_x \sum_{i=1}^N \sum_{j=1}^N \sigma_{ij} w_i w_j$	$\frac{1}{2} x^T P x + q^T x$	$P = 2\Sigma, q = 0$

$$\sum_{i=1}^n \mu_i v_i = \text{min row 0 of } A \quad b$$

$$\sum_{i=1}^n x_i = 0 \quad \text{row 1 of } A \quad c$$

$$w_i \geq 0 \quad \text{row 2 - n of } A \quad l=0 \quad u=+\infty$$

Some QSDP advantages

One factorization for the whole fraction. In the fraction only b changes so the factorization is faster

Warm start. The previous sol is used for the next problems

Accuracy effing. It allows us to define the number of iterations and the tolerance. Since the covariances are 10^{-3} the tolerance decided is 10^{-6}

Instances

The instances created in this edition correspond to OR-Library built by Cheng, Mark, Beasley and Sharaiha (2000).

These instances was the following sets
Distance Index Market N μ_i x_i

port 2	Hong Kong	:	:
:	:	:	:
port 5	Nikko	Japan	225

furthermore, each instance has the optimal portfolio at port of X . text which contains the proportion of assets with highest expected return to each risk

The solution proposed

for each

table